

OPERATING SYSTEMS (CS304)

Scheduling Algorithms (FCFS, SJF, RR)

Learning Objectives

Understand CPU scheduling, scheduling criteria, FCFS, SJF, and Round Robin algorithms, along with numerical examples and performance measures.

Introduction to CPU Scheduling

CPU scheduling is the process of selecting a process from the ready queue and allocating CPU time to it. Efficient scheduling improves CPU utilization and system performance.

Need for CPU Scheduling

- Maximize CPU utilization
- Improve throughput
- Reduce waiting time
- Reduce turnaround time
- Improve system responsiveness

Scheduling Criteria

1. CPU Utilization
2. Throughput
3. Turnaround Time
4. Waiting Time
5. Response Time

First Come First Serve (FCFS)

FCFS is the simplest scheduling algorithm. The process that arrives first gets CPU access first.

Advantages of FCFS

Simple implementation, low overhead, easy to understand.

Disadvantages of FCFS

May cause long waiting times and the convoy effect.

Shortest Job First (SJF)

SJF selects the process with the smallest CPU burst time for execution.

Advantages of SJF

Provides minimum average waiting time among scheduling algorithms.

Disadvantages of SJF

Difficult to predict exact burst times and may cause starvation of long jobs.

Round Robin (RR)

Round Robin allocates CPU to each process for a fixed time quantum in cyclic order.

Advantages of RR

Fair scheduling, suitable for time-sharing systems, and improved response time.

Disadvantages of RR

Too small quantum increases context switching overhead; too large quantum behaves like FCFS.

Numerical Example

Processes: P1 = 5 ms

P2 = 3 ms

P3 = 8 ms

FCFS Gantt Chart: P1 | P2 | P3

Waiting Times: P1 = 0, P2 = 5, P3 = 8

Comparison of Scheduling Algorithms

FCFS: Simple, may cause convoy effect.

SJF: Minimum average waiting time.

RR: Best for interactive systems.

Exam Important Questions

1. Define CPU Scheduling.
2. Explain FCFS Scheduling.
3. Explain SJF Scheduling.
4. Explain Round Robin Scheduling.
5. Compare FCFS, SJF, and RR.
6. Solve numerical problems based on CPU Scheduling.

Summary

CPU Scheduling is essential for efficient resource utilization. FCFS, SJF, and RR are widely used scheduling algorithms, each with unique advantages and limitations.